

Thought Experiments through the First Three Chapters of “Foundations of Quantum Theory”

—From the Identification Wall in Measurement to the Algebra of Observables

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Abstract

Against the background of the framework of complex Hilbert spaces, observables, and measurement developed in Chapters 1 to 3 of Akira Shimizu’s *New Edition: Foundations of Quantum Theory* [1], this paper organizes five stepwise thought experiments to unify the treatment of measurement precision, the uncertainty relation, wave packets, quantum correlations, and the algebra of observables. The starting point is the “identification wall” of classical measurement theory, and the destination is the algebraic picture in which “physical quantities satisfying an uncertainty relation are different projections extracted from the same wave packet.” The core reading of the present paper is a semiclassical and geometric interpretation that positions the invariants in phase space as symplectic capacities, exemplified by de Gosson’s quantum blob [9]. The present paper does not contradict the mathematical system of standard quantum theory; rather, it offers one reading that supplements its conceptual outlook.

Thought Experiment I: The Identification Wall in Measurement

Setting

Suppose a physical quantity A takes real values. We measure the distance L between two points A and A using a measure based on the concept of distance, which returns only discrete values with a minimum spacing of ϵ .

At each measurement, one of the two adjacent lattice points $\{k, (k+1)\}$ bracketing the true value is returned with probability depending on the position r of the true value within the cell:

$$P(\text{up}) = r/\epsilon, P(\text{down}) = 1 - r/\epsilon$$

(position-dependent dithering).

Behavior of the N-fold Average

When $A = A$, the single-shot distance $D = \epsilon \lfloor A/\epsilon \rfloor$ takes values $\{0, \epsilon, 2\epsilon, \dots\}$. Writing the within-cell position of the true value as $q = r/\epsilon \in [0, 1]$, in general:

- $P(D = 0) = q^2$
- $P(D = \epsilon) = 2q(1-q)$
- $P(D = 2\epsilon) = (1-q)^2$

Hence $E[D] = \epsilon q$, $\text{Var}[D] = \epsilon^2 q(1-q)$, $\lim_{N \rightarrow \infty} D/N = 0$.

In particular, at the symmetric position $q = 1/2$, the probability distribution is $\{1/4, 1/2, 1/4\}$, $\text{Var}[D] = \sigma^2/2$, and $\sigma_N = \sigma/\sqrt{2N}$.

As $N \rightarrow \infty$, $\sigma_N \rightarrow 0$, and $L_N \rightarrow L$ with probability 1.

Condition for Distinguishability

For A and A' to be recognized as two distinct objects, the two must belong to different cells. If they fall within the same cell, the measure cannot distinguish “one point or two,” and the quantity “distance between two points” cannot be constituted.

Therefore the condition for the problem of measuring L to be well-posed is

$L > \sigma$.

The region $L < \sigma$ is not “buried in measurement error”; it is a region in which the very concept of “two points” is not defined.

Conclusion

When a measure based on distance is used to measure distance, measurement at scales below the measure’s spacing is **undefined** prior to any invocation of the uncertainty principle. Functions not as a resolution but as the threshold of individuation.

Within the range $L > \sigma$, arbitrary precision is attainable by repeated measurement regardless of the absolute value of L .

The present argument holds in general for any measure based on the concept of distance and has a structural analogy with, but is positioned independently of, the discussions of minimum length at the Planck scale in quantum gravity [10].

More rigorously: rather than the ontological statement “no two points exist below σ ,” the appropriate formulation is “no projection operator distinguishing two points exists in the observable algebra generated by a σ -step measure.” The present discussion can be made consistent with the framework of POVM and Kraus operators in generalized measurement theory, where it appears as a special case of coarse-grained measurement.

Note also that when A and A' are externally pre-labeled as distinct objects, the mean sub-cell difference can be estimated by repeated measurements (the $N \rightarrow \infty$ convergence via position-dependent dithering). The “undefinability” referred to in the present paper applies to the case where no operation exists within the observing system itself to individuate two objects.

Thought Experiment II: The Locus of Fluctuation

Setting

Consider the reverse situation. The measure has infinite precision and returns its input value unchanged. The physical quantity A itself, however, fluctuates about its true value A' according to

$$A_i = A' + \epsilon_i, \epsilon_i \sim \text{Uniform}[-\sigma/2, +\sigma/2].$$

We assume $L > \sigma$.

N-fold Average

- $E[L_N] = A'$
- $\text{Var}[L_i] = \sigma^2/12$
- $\sigma_N = \sigma/(12N) \rightarrow 0$

Comparison with Thought Experiment I

Instantiating both cases with true value 7.7 and spacing $\Delta = 1$, each yields the same empirical distribution: out of 100 measurements, roughly 30 return 7 and 70 return 8, with the average converging to 7.7.

	Fluctuation on the measure side	Fluctuation on the quantity side
Single-shot output space	Discrete $\{k, (k+1)\}$	Continuous
Probability distribution	Position-dependent Bernoulli	Additive uniform
$N \rightarrow \infty$ convergence target	True value	True value
Distinguishability from observed data	Impossible	

Two-sided Fluctuation

When fluctuations Δ_q (width Δ_q) on the quantity side and Δ_m (width Δ_m) on the measure side coexist independently:

- Observed value: $L_i = A' + \Delta_m + \Delta_q$
- $\sigma_N = (\text{Var}[\Delta_m] + \text{Var}[\Delta_q]) / N \rightarrow 0$

The $1/N$ decay persists, but separating Δ_m and Δ_q from a single observed series requires additional assumptions external to the observation (independent calibration of the measure, time-series decomposition, prior knowledge of distribution shapes, etc.).

Conclusion

Defining the observed fluctuation magnitude as σ_{obs} , one cannot determine from observed data alone whether it originates from the quantization width of the measure, from the fluctuation of the quantity itself, or from a combination of both.

Rather than “convergence to the objective true value,” the precise statement is “convergence to the effective true value defined by the observer,” and the $1/N$ decay applies rigorously in this sense.

More rigorously: “cannot be determined from observed data alone” means “under the condition of a single observed series, a single measurement setup, and no prior calibration.” If the apparatus Hamiltonian is known, system-side and environment(measure)-side fluctuations can in principle be distinguished via open-system decoherence theory. The present paper’s claim is unidentifiability from observation alone, which connects to the observer-knowledge-dependent epistemological aspect (QBism-type interpretations).

Thought Experiment III: Wavelength Representation of Momentum and the Uncertainty Relation

Setting

We extend the object of Thought Experiment II to a quantum-theoretic wave packet. For position x , consider the center position (expectation value $\langle x \rangle$) x and the spread x of the wave packet. Here “center position” is merely a label of the packet distribution; we do not take the position that there exists an objective true value external to observation.

Similarly for momentum p , consider the center position (expectation value $\langle p \rangle$) p and the spread p .

Rewriting via Wavelength

From the de Broglie relation [2]:

$$p = \frac{h}{\lambda} = \hbar k, \quad k = \frac{2\pi}{\lambda}$$

the momentum fluctuation is

$$|\delta p| = \frac{h}{\lambda^2} |\delta \lambda| = \hbar |\delta k|.$$

(The differential relation $dp/d\lambda = -h/\lambda^2$ has a sign, but we take absolute values for fluctuation magnitudes.)

Rewriting the uncertainty relation $\Delta x \Delta p \geq \hbar/2$ [7] in terms of the wavenumber:

$$\Delta x \cdot \Delta k \geq \frac{1}{2}.$$

Observation

In this form, Planck’s constant disappears from both sides. What remains is a mathematical inequality of Fourier transforms concerning the product of position and wavenumber.

Planck’s constant functions as a dimensional conversion factor connecting the unit of position (m) and the unit of momentum ($\text{kg} \cdot \text{m/s}$). A similar view that treats Planck’s constant not as an observable quantity but as a convention of human unit-choice has been proposed by Ralston [8].

Qualitatively:

- Pinning position precisely narrowing the position wave packet broadening the wavelength distribution making momentum vague
- Pinning momentum precisely narrowing the wavelength to one spreading over all of space making position vague

Packet form	Spatial spread	Wavelength distribution
Delta function	$x = 0$	All wavelengths
Rectangular pulse	Finite	sinc function
Pure sine wave	All of space	Single wavelength

Packet form	Spatial spread	Wavelength distribution
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Conclusion

In the wavenumber representation, the position–momentum uncertainty relation becomes $xk \geq 1/2$, described as a lower bound on the product of conjugate quantities in Fourier transforms. Planck’s constant appears as a dimensional conversion factor.

More rigorously: $\hbar$ is the quantum of action appearing in the canonical commutation relation $[x, p] = i\hbar$, and has physical content independent of the fact that $\hbar = 1$ can be set in natural units. The present paper’s claim is restricted to the fact of rewriting: the transition to the wavenumber representation rewrites the canonical commutation relation as $[x, k] = i$, and allows the uncertainty relation to be read as a geometric inequality of wavenumber conjugacy. “ $\hbar$ disappears” means it disappears from the surface of the expression; it does not deny the physical role of $\hbar$ as the quantum of action.

Thought Experiment IV: Quantum Correlations as a Composite Wave Packet

Setting

Consider a state localized in two spatial regions A and B with correlations mediated by a shared conserved quantity (momentum, energy, etc.). In standard descriptions, this system is treated as a “two-particle entangled state” [3, 4, 5].

We attempt to describe this instead as “a state in which a single composite wave packet is localized in two regions in space.”

String Analogy

When a long, taut string undergoing baseline vibration has one end suddenly fixed, a soliton-like deformation appears at the opposite end. This is a result of the entire string obeying conservation laws as a single system; no information has propagated from one end to the other.

Geometry of the Composite Wave Packet

- The composite wave packet occupies one area element in phase space
- In configuration space it has localized peaks in two regions A and B
- A local interaction occurs on the A side, contracting x_A
- Due to the Robertson inequality [6] and Robertson–Schrödinger-type covariance constraints, a reduction of spread along one projection direction induces compensating changes in the conjugate direction or covariance component, broadening p_A
- To satisfy the geometric constraints from the symplectic capacity of the entire composite packet, the spread of the wave packet on the B side also undergoes geometric deformation

However, the “deformation on the B side” referred to here is a change in the description of the post-measurement state *conditioned on the measurement result on the A side*, and does not mean that the unconditioned reduced density matrix on the B side changes under local operations on the

A side. This distinction maintains consistency with the no-signalling theorem of standard quantum theory.

In this description, no information transfer from A to B is introduced. The observed correlation is a geometric consequence reflecting the state of the entire system, conserved quantities and symmetries, and the symplectic-capacity constraints.

The True Nature of the Geometric Invariant

What constrains the geometry of the composite system is not individual wavelengths or positions, but the **symplectic capacity** formed by the product of the spreads of conjugate physical quantities (an area element in phase space):

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}, \quad \Delta A \cdot \Delta B \geq \frac{1}{2} | \langle [A, B] \rangle |.$$

The left side has the dimension of area; the inequality means a lower bound on the area. This is not a simple conservation law of xp , but should be understood as a geometric constraint by the symplectic capacity, which is invariant under canonical transformations.

Strictly, in multi-degree-of-freedom systems, the symplectic capacity is not defined as a simple total phase-space volume or as the product of component uncertainty products, but as a symplectic invariant concerning projections onto canonical 2-planes (Gromov’s non-squeezing theorem [15]). In the present paper, we use the xp -type area element as its intuitive shadow that conveys the essence.

Conclusion

Quantum correlations are describable as “a state in which one composite wave packet is localized across multiple regions in space.” The observed correlations are geometric consequences reflecting the state of the entire system, conserved quantities and symmetries, and the symplectic-capacity constraints; this description does not require information transfer between spatially separated systems.

More rigorously: the composite wave packet picture of the present paper is an attempt to read a single state vector on the tensor-product Hilbert space of standard quantum theory through the geometric intuition of phase space. The string-soliton analogy provides an intuitive presentation of the geometric structure of conservation laws; it is not a complete re-derivation of the quantitative predictions of Bell-inequality violations, for example the maximal violation 22 of the CHSH inequality (the Tsirelson bound). Those quantitative predictions follow from the algebraic structure of quantum theory independently of any classical wave analogy. The present description’s “no information transfer required” property is consistent with the no-signalling theorem of standard quantum theory (partial trace of the density operator, invariance under local operations).

Thought Experiment V: The Algebra of Observables

Preface on the validity range of metaphors: In this section, we use the intuitive terms “wave packet,” “area element,” and “projection.” These are valid as semiclassical and geometric metaphors for pure states in position-momentum systems (infinite-dimensional, continuous spectrum). For spin systems and finite-dimensional systems, they should be reread as “wave packet = point on projective

Hilbert space, state direction” and “area element = geometry of the non-commutative algebra of projection operators.” Each “More rigorously” section at the end of the thought experiments supplements the strict mathematical correspondences of these expressions.

Setting

We list several pairs of physical quantities that satisfy the uncertainty relation $AB = |\langle[A,B]\rangle|/2$:

- Position x momentum p
- Spin S_x S_y S_z
- Polarization (H/V D/A R/L)
- Angular momentum L_x L_y L_z
- Energy time (note: the energy-time uncertainty relation has a different mathematical status from Robertson-type uncertainty relations between position-momentum or spin components, because time is not generally defined as a self-adjoint operator in standard quantum mechanics)

All of these share a common structure.

Common Structure

Pair of quantities	Shared structure	Role of distinct operators
x p	Area element in phase space	Direct coordinate Fourier coordinate
S_x S_y S_z	Area element in spin phase space	Projection onto distinct axes
Polarization basis	Area element of polarization plane	Projection onto distinct bases
L_x L_y L_z	Angular-momentum phase space	Projection onto distinct axes
E t	Energy–time phase space	Distinct representations

Unified Picture

In a complex Hilbert space:

Formal object	Geometric interpretation
Eigenvector $ \rangle$	Wave packet (one area element in phase space)
Operator $\hat{A}$	Projection extracting a real value from the area element
Eigenvalue a	Result of the projection
Non-commutativity $[\hat{A}, B] \neq 0$	Two projections viewing the same area element from different directions

Conclusion

Physical quantities satisfying an uncertainty relation are described as distinct projections of the same wave packet (eigenvector). The product of non-commuting projections is bounded below by the area element of the wave packet (Robertson inequality).

In the geometric reading of the present paper, a physical quantity is understood as an observable that extracts a real-valued distribution from a state, and its value is obtained through the measurement basis or spectral projection. A wave packet is interpreted, in the semiclassical phase-space representation, as an area element whose spread is constrained by Robertson–Schrödinger-type covariance constraints and by the symplectic capacity. The projections of observables do not take arbitrary independent values, but are correlated under the geometric invariants of the same state.

The area-element picture of the present paper is consistent with de Gosson’s *quantum blobs* [9] — the smallest phase-space units saturating the Robertson–Schrödinger inequality, measured by symplectic capacity, that is, an area-type invariant on canonical 2-planes. The present paper can be regarded as a reading of Chapter 1 of Shimizu’s textbook that reconstructs this mathematically heavy symplectic-geometric framework in the intuitive form of thought experiments. Related geometrical formulations include the Kähler-geometric formulation on projective Hilbert space [11] and the algebraic approach in which the algebra of observables is taken as primary [12].

More rigorously: “operator = projection” risks confusion with *projection operator* as a special concept; observables are *self-adjoint operators*, and in projective measurements values are extracted via spectral projections (for generalized measurements, POVMs). “Eigenvector = wave packet” is an effective metaphor for position-momentum systems, but for spin systems and finite-dimensional systems it should be interpreted as “a state direction, a point in projective Hilbert space.” The phase-space area-element picture of the present paper is a semiclassical and geometric interpretation; rigorous formulations purely on Hilbert space are given by Weyl quantization, Wigner functions [13], Moyal brackets [14], and geometric quantization.

Note that the Robertson inequality itself is a lower-bound condition: for position and momentum in infinite-dimensional Hilbert space, there is no general absolute upper bound on $(x)(p)$ (in a free-particle Gaussian wave packet, x can grow with time). However, in finite-dimensional systems, bounded observables, and state-dependent contexts, *reverse uncertainty relations* with upper-bound-type uncertainty relations have been studied in recent years. The geometric-invariant claim of the present paper is most accurately read not as a simple conservation law of xp , but as the invariance of the symplectic capacity under canonical transformations (de Gosson’s quantum blob [9], Gromov’s non-squeezing theorem [15]).

Summary

Through the five thought experiments, the following structure emerged:

1. **The identification wall** (I) — is not a resolution but a threshold of individuation. L is a prerequisite for distance measurement.
2. **Indistinguishability of the locus of fluctuation** (II) — on the measure side and on the quantity side cannot be distinguished from observed data.
3. **Uncertainty relation in the wavenumber representation** (III) — $xk = 1/2$ is a mathematical fact of Fourier transforms. $\hbar$ appears as a unit conversion factor.
4. **Quantum correlations as a composite wave packet** (IV) — Described as the spatial division of a single wave packet, the correlations between spatially separated systems can be explained without assuming information transfer.
5. **The algebra of observables** (V) — Physical quantities satisfying an uncertainty rela-

tion are distinct projections of the same wave packet. The area element in phase space (symplectic capacity) acts as a geometric invariant that imposes constraints under canonical transformations.

These are consistent with the framework of complex Hilbert spaces, observables, and measurement introduced in Chapters 1 to 3 of Shimizu's *New Edition: Foundations of Quantum Theory*, and present one reading that provides a conceptual outlook on quantum theory. The present paper does not modify the mathematical predictions of standard quantum theory; it confines itself to offering a geometric interpretation of each concept.

Open Problems and Future Work

To develop the thought experiments of the present paper into a fully mathematical formulation, the following routes are possible:

1. **Formulating Thought Experiments I and II as POVM / coarse-grained measurement:** rigorously treating the identification wall and the indistinguishability of fluctuation locus within the framework of Kraus operators and generalized measurement, and clarifying the relation to decoherence theory when the apparatus Hamiltonian is known.
2. **Formulating Thought Experiment III via Fourier analysis and the Robertson inequality:** deriving the position-wavenumber conjugacy from the Plancherel theorem and clarifying $\hbar$'s role as an action-dimensional coefficient.
3. **Formulating Thought Experiment IV via tensor-product space, partial trace, and the no-signalling theorem:** refining the composite wave packet picture as the geometry of reduced states of density operators, and quantitatively deriving the maximal CHSH violation 22 (the Tsirelson bound) for Bell-inequality violations.
4. **Formulating Thought Experiment V via C^* -algebras / von Neumann algebras / geometrical quantum mechanics:** a self-contained description of the algebra of observables, and explicating its correspondence with the Kähler manifold structure on projective Hilbert space [11].
5. **Explicating correspondences with de Gosson's quantum blobs [9], Wigner functions [13], and Moyal brackets [14]:** connecting to phase-space quantum mechanics via symplectic capacity, the Wigner quasi-probability distribution, and the Moyal star product.

These are research programs that go beyond the observational framework of the present paper and should be treated as independent successor works.

Priority outlook: Of the above, the most natural successor candidates are (1) the formulation of Thought Experiments I and II as POVM / coarse-grained measurement, and (5) the explication of correspondences with de Gosson's quantum blobs. The former directly connects to standard quantum measurement theory and rigorously formalizes the starting point of the present paper (the identification wall and the indistinguishability of the locus of fluctuation); the latter positions the core picture of the present paper (the phase-space area element) within the existing framework of symplectic geometry. Items (2), (3), and (4) can then be expanded successively on this foundation.

Acknowledgment

The present thought experiments were progressively organized through dialogue with AI. Verbatim records including the trial-and-error process are publicly available in the ai-chat-logs-open repository (/ (6)*, (7)*).

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